TAUTNESS OF A COMPOSITE IMPLIES TAUTNESS OF ITS SECOND FACTOR

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Allowed standard prerequisites (editorial).

- Tautness of the composite map, giving openness of the closures of the extensions of open ideals of A. Same chapter, the definition of a taut map, in force in the statement and used at proof lines 1377-1382. Supplies openness of the closures used in both parts of the proof.
- The inverse image of an open ideal, and the closure of the extension of an ideal. Standard, used at proof lines 1375-1384. Constructs the ideals of B whose extensions realise the fundamental system of C.

Lemma 0.1. *Let $A \to B$ and $B \to C$ be continuous homomorphisms of linearly topologized rings. If $A \to C$ is taut, then $B \to C$ is taut.*

Proof. Let $J \subset B$ be an open ideal with inverse image $I \subset A$. Then the closure of JC contains the closure of IC. Hence this closure is open as $A \to C$ is taut. Let I_λ be a fundamental system of open ideals of A. Let K_λ be the closure of $I_\lambda C$. Since $A \to C$ is taut, these form a fundamental system of open ideals of C. Denote $J_\lambda \subset B$ the inverse image of K_λ. Then the closure of $J_\lambda C$ is K_λ. Hence we see that the closures of the ideals JC, where J runs over the open ideals of B form a fundamental system of open ideals of C. □

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